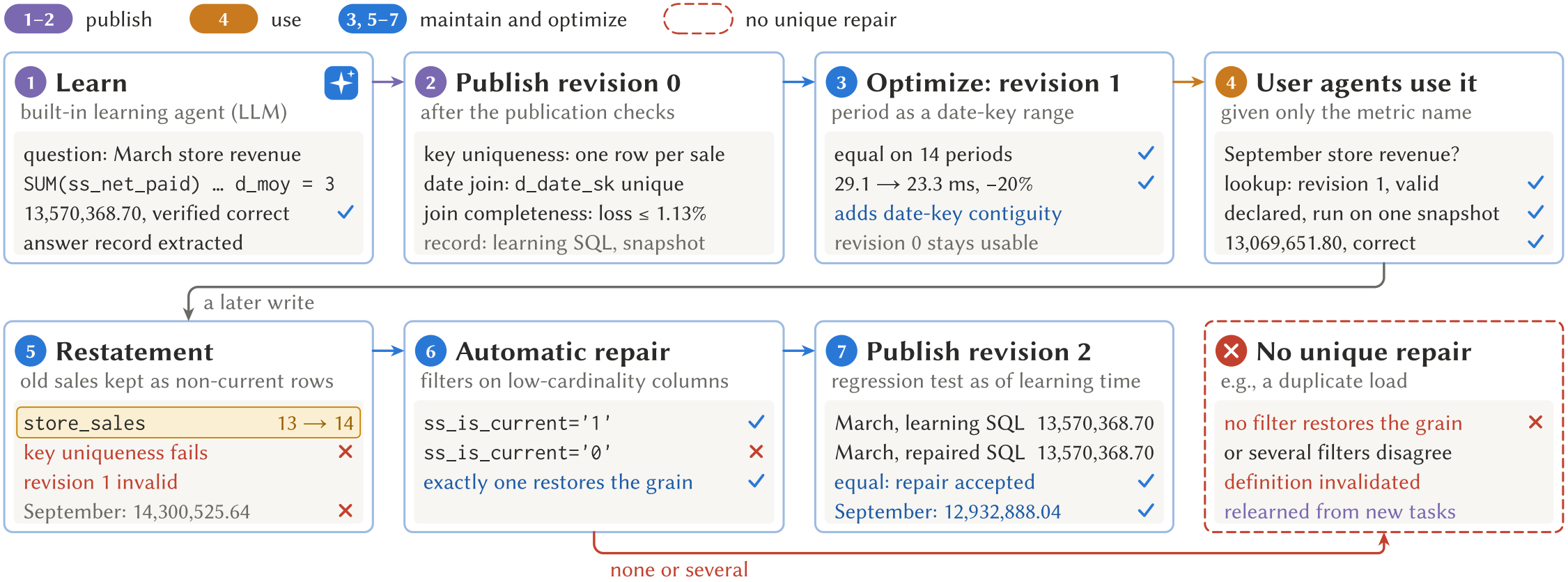

1–2 publish
4 use
3, 5–7 maintain and optimize
no unique repair
1 Learn
built-in learning agent (LLM)
question: March store revenue
SUM(ss_net_paid) … d_moy = 3
13,570,368.70, verified correct
answer record extracted
2 Publish revision 0
after the publication checks
key uniqueness: one row per sale
date join: d_date_sk unique
join completeness: loss ≤ 1.13%
record: learning SQL, snapshot
3 Optimize: revision 1
period as a date-key range
equal on 14 periods
29.1 ⟶ 23.3 ms, −20%
adds date-key contiguity
revision 0 stays usable
4 User agents use it
given only the metric name
September store revenue?
lookup: revision 1, valid
declared, run on one snapshot
13,069,651.80, correct
a later write
5 Restatement
old sales kept as non-current rows
store_sales 13 ⟶ 14
key uniqueness fails
revision 1 invalid
September: 14,300,525.64
6 Automatic repair
filters on low-cardinality columns
ss_is_current='1'
ss_is_current='0'
exactly one restores the grain
7 Publish revision 2
regression test as of learning time
March, learning SQL 13,570,368.70
March, repaired SQL 13,570,368.70
equal: repair accepted
September: 12,932,888.04
No unique repair
e.g., a duplicate load
no filter restores the grain
or several filters disagree
definition invalidated
relearned from new tasks
none or several